EMPLOYEE ATTRITION PREDICTION USING MACHINE LEARNING

XYlofy AI Internship Program – Week 2
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EXECUTIVE SUMMARY

Employee attrition is a significant challenge for organizations because employee turnover increases recruitment costs, onboarding expenses, productivity loss, and operational disruption. This project developed a Machine Learning solution capable of identifying employees who are likely to leave the organization before attrition occurs.

The IBM HR Analytics dataset containing 1,470 employee records was analyzed. Employee demographics, compensation, overtime behavior, satisfaction levels, work-life balance, and career progression factors were evaluated to understand attrition patterns and support HR decision-making.

Three machine learning models were developed and compared: Logistic Regression, Random Forest, and Gradient Boosting. Gradient Boosting achieved the highest overall predictive performance with an ROC-AUC score of 0.804 and an accuracy of 85%.

DATASET OVERVIEW

Total Employees	1470
Total Features	35
Attrition Cases	237
Retention Cases	1233
Attrition Rate	16.12%
Missing Values	0
Train-Test Split	80:20

KEY BUSINESS FINDINGS

• Sales Department recorded the highest attrition rate (20.63%).

• Sales Representatives experienced the highest employee turnover (39.76%).

• Employees working overtime showed attrition of 30.53% compared to only 10.44% among employees without overtime.

• Lower-income employees demonstrated significantly higher attrition risk.

• Poor work-life balance was strongly associated with employee turnover.

• Early-career employees were more likely to leave than long-tenured employees.

MODEL PERFORMANCE

Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	0.75	0.35	0.62	0.798
Random Forest	0.84	0.57	0.09	0.781
Gradient Boosting	0.85	0.59	0.21	0.804

Gradient Boosting achieved the strongest overall predictive performance. However, Logistic Regression demonstrated the highest recall for identifying employees at risk of leaving, making it valuable for proactive HR interventions.

TOP FACTORS DRIVING ATTRITION

• Monthly Income

• Age

• Total Working Years

• OverTime

• Number of Companies Worked

• Stock Option Level

• Years With Current Manager

• Daily Rate

• Environment Satisfaction

• Job Involvement

HR RECOMMENDATIONS

• Reduce excessive overtime through better workload planning and workforce allocation.

• Prioritize retention initiatives for Sales Representatives and Sales employees.

• Improve compensation competitiveness for high-risk employee groups.

• Strengthen employee engagement and satisfaction programs.

• Provide structured mentoring and career development opportunities for early-career employees.

CONCLUSION

Employee attrition can be predicted using employee demographics, compensation, work environment, and career progression factors. Monthly Income, Age, Total Working Years, and Overtime emerged as the strongest predictors of employee turnover. The insights generated through this analysis can help HR departments reduce attrition, improve employee satisfaction, and support long-term workforce stability.